Junkun Liu

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Education

Johns Hopkins University Baltimore, MD

B.S. in Applied Mathematics & Statistics and Computer Science | GPA: 3.74/4.00 *Expected May 2028*

Relevant Coursework: Machine Learning & Deep Learning, NLP: Self-Supervised Models, Software System Design, Computer System Fundamentals, Data Structures & Algorithms, Probability, Optimization

Experience & Projects

LaunchStack – *Tech Lead* *Jun 2026 – Present*

- Lead technical design and engineering delivery for the **Founder Operating System**, translating product goals into system architecture, implementation plans, and cross-team priorities.
- Designed the **Vantage** decision-support pipeline, connecting role-specific context and evidence retrieval to cited AI reports, human review, and tracked commitments.
- Lead **code/PR reviews** across frontend and AI workstreams, reviewing implementation quality and alignment with system design before integration.

Aftershock – *Project Lead, HopHacks 2026* *Sep 2026*

- Won **1st place among 27 teams** in the Bloomberg Most Philanthropic Hack track; led a **4-person team** to a working earthquake-response demo in **36 hours**.
- Prototyped AI-assisted incident fusion from conflicting reports and **dynamic rerouting** for rescue teams over real road networks.

StepFun – *Algorithm Engineer Intern* *Jun 2025 – Aug 2025*

- Evaluated **LLM-based web agents** for multi-step navigation, tool coordination, and dynamic-DOM interaction.
- Investigated browser-agent failures involving navigation loops, ambiguous elements, tool misuse, and unstable action sequences.

Research & Publications

Mystra: Dynamic Taint Analysis – *Second Author,* ***ICSE 2027 (Accepted)*** *Jun 2026 – Jul 2026*

- Built **reproducible full-application CVE validations** tracing tainted input to DOM and code-execution sinks; analyzed propagation and source-to-sink coverage limitations.
- Supported cross-runtime porting and evaluation of a declarative **shadow-VM taint-analysis system** spanning CPython, SpiderMonkey, and Chromium.

Genesis Mission – *Undergraduate Researcher, Prof. Ziyang Li* *Sep 2026 – Present*

- Contribute to research on **AI-generated scientific computing software**, focusing on reliable GPU-accelerated code for computational-physics simulations.
- Build **reproducible CPU/GPU validation workflows** by reconstructing simulation workloads, configuring CUDA environments, and checking generated code against trusted numerical references.

Leadership & Teaching

Johns Hopkins PILOT Program – *Head PILOT Leader* *Jul 2026 – Present*

- Promoted after one year; lead weekly **Honors Discrete Mathematics** proof-solving sessions for **10–15 students**.

Johns Hopkins University – *Teaching Assistant, Mathematical Foundations of CS* *Fall 2026 – Present*

- Support instruction in **proofs and discrete structures**; selected after earning an A+ in the course.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, SQL

AI/ML: LLM Agents & Evaluation, RAG, Structured Outputs, Embeddings, PyTorch, scikit-learn

Systems/Tools: Linux/WSL, CUDA, SQLite, PostgreSQL, Docker, Git, React, Next.js, Node.js, AWS